Representation Learning and Orthogonal Subspace Fusion for Generalizable Image Retrieval

Theoretical Problem Deconstruction

Image retrieval is frequently reduced to a surface-level operational task: querying a database with an input image and returning a ranked list of visually similar output images based on a defined distance metric. However, to engineer a next-generation retrieval architecture, one must deconstruct this operational view and address the fundamental theoretical challenges underlying the system. At its absolute core, content-based image retrieval (CBIR) is a complex intersection of deep representation learning, invariant feature extraction, and high-dimensional metric learning.[1] The true objective of the system is not merely to compare pixels, but to map high-dimensional, raw sensory inputs into a lower-dimensional, continuous, and highly structured metric space. In this target latent space, the geometric distance between any two feature vectors must accurately and robustly reflect their high-level semantic similarity, independent of environmental noise, perspective shifts, or intra-class variations.[2]

Achieving this optimal mapping requires overcoming the widely documented "semantic gap." This gap defines the fundamental disconnect between the low-level sensory data captured by a machine—specifically the discrete matrices of pixel intensities—and the high-level conceptual abstractions, semantic meanings, and contextual relationships perceived by human cognition.[2] Bridging this gap necessitates representations that exhibit simultaneously high inter-class separability and tight intra-class clustering. Furthermore, the learning problem demands that the system intrinsically acquire critical invariances. A robust retrieval model must learn translation, scale, and illumination invariances to generalize across diverse environments, while simultaneously preserving the discriminative equivariant features that rigidly define specific fine-grained object identities and categories.[3] Achieving this delicate balance between invariance to noise and sensitivity to identity requires an architectural design that rigorously addresses the specific theoretical and empirical failure modes of historical and contemporary representation learning paradigms.

Diagnosis of Representation Failure Modes

The historical evolution of computer vision features reveals a sequence of distinct failure modes across different representational paradigms, spanning from early handcrafted descriptors to contemporary deep convolutional embeddings. A rigorous diagnosis of these vulnerabilities is required to justify the synthesis of a novel, hybrid architecture.

Handcrafted Features: Local Binary Patterns (LBP)

Local Binary Patterns (LBP) and similar traditional handcrafted features operate by deterministically thresholding the neighborhood of each pixel against the central pixel, treating the resulting Boolean array as a binary number to capture micro-textural structural information.[1] While these algorithms are celebrated for their computational efficiency and partial invariance to monotonic grayscale transformations, they suffer from severe and insurmountable theoretical limitations in the context of modern semantic retrieval.[5]

The primary failure mode of LBP is its absolute inability to bridge the semantic gap. LBP exclusively captures localized, low-level visual statistics, such as micro-edges, corners, and isolated local textures, without possessing any hierarchical, compositional mechanism to assemble these low-level primitives into high-level object semantics.[2] Furthermore, LBP architectures are fundamentally rigid. Because the feature extraction algorithm is fixed by human design rather than optimized via gradient descent for a specific data distribution, the resulting representations cannot adapt to the underlying manifold of the dataset.[2] To capture multi-scale spatial information, LBP must be independently computed across an expanding range of radii and neighborhoods, leading to an exponentially growing, highly sparse, and high-dimensional feature vector that inevitably triggers the curse of dimensionality during the similarity matching phase.[7] Finally, LBP is highly susceptible to imaging noise, sensor artifacts, and variations in spatial resolution. These environmental perturbations corrupt the uniform binary patterns that are mathematically necessary for accurate and consistent texture description, leading to catastrophic retrieval failures in unconstrained, real-world environments.[3]

Shallow Neural Networks and Multi-Layer Perceptrons

Shallow Neural Networks, conventionally implemented as Multi-Layer Perceptrons (MLPs), represent the earliest attempts to automatically learn representations directly from data distributions, bypassing the limitations of human-engineered features. However, their failure in complex, real-world vision tasks is profoundly rooted in an absolute lack of spatial inductive bias.[8]

In a standard MLP architecture, a two-dimensional image matrix must be flattened into a one-dimensional vector before processing. This fundamental operation renders the network architecture invariant to the permutation of input pixels, meaning the model entirely and irreversibly discards the inherent two-dimensional topological structure, spatial proximity, and geometric relationships of the image.[8] Consequently, MLPs suffer from an absolute lack of translation equivariance. Because these networks rely on dense, fully connected weight matrices without spatial weight sharing, they are forced to independently relearn feature detectors for every single spatial location across the image array. A vertical edge detector learned in the top-left quadrant of the image cannot be applied to the bottom-right quadrant,

leading to massive representational redundancy.[8] To compensate for this profound lack of structural inductive bias, MLPs require an exponential scaling in both trainable parameters and training data volume to achieve even basic spatial recognition.[8] This overparameterization makes shallow networks highly prone to severe overfitting on small datasets, yielding remarkably poor out-of-distribution (OOD) generalization and rendering them practically obsolete for unconstrained image retrieval tasks.[8]

Convolutional Neural Network (CNN) Embeddings

While Convolutional Neural Networks (CNNs) effectively resolved the spatial inductive bias problem of MLPs through the introduction of local receptive fields and weight sharing across the spatial domain [9], they introduce their own modern, subtle, and highly impactful failure modes in retrieval environments.

A critical vulnerability of CNN embeddings is the extensively documented texture bias versus shape bias dilemma. Empirical analysis consistently demonstrates that standard CNN architectures, including deep residual networks (ResNets), possess a remarkably strong bias toward classifying and retrieving images based on localized textures rather than global structural contours or semantic shapes.[11] Because the convolution operation inherently aggregates local pixel statistics, CNNs often prioritize high-frequency textural patterns. If a query image contains an object with an unusual, out-of-distribution texture but a perfectly standard structural shape, the CNN embedding will frequently fail to retrieve structurally similar objects, opting instead to retrieve semantically unrelated objects that share the anomalous texture.[11]

Furthermore, CNNs suffer from a systematic loss of high-frequency spatial information. The repeated utilization of spatial downsampling layers, such as max pooling or strided convolutions, is architecturally necessary to achieve spatial invariance and expand the receptive field. However, this downsampling inherently discards critical high-frequency spatial details, phase information, and fine-grained structural cues that are often strictly necessary for exact instance-level retrieval.[13] Finally, deep CNN embeddings exhibit severe vulnerability to domain shift. Because they are highly parameterized, they tend to overfit to the specific statistical regularities, lighting conditions, and perspective projections of their training domain (such as the natural, horizontally projected images in ImageNet or CIFAR-10). When deployed in cross-dataset retrieval scenarios involving novel sensory modalities or perspectives, the metric space collapses, leading to severely degraded generalization.[14]

The Geometric Flaws of Naive Feature Fusion

To theoretically compensate for the isolated weaknesses of individual modalities—such as combining the texture sensitivity of LBP with the semantic depth of a CNN—traditional retrieval systems frequently resort to multi-modal feature fusion.[1] However, naive fusion methodologies,

such as direct vector concatenation ($f_i = [f_{CNN} |

| f_{Proposed}]$) or fixed scalar linear combinations ($f_i = \lambda f_{CNN} + (1 - \lambda) f_{Proposed}$), present critical theoretical and geometric flaws.[1]

The primary failure mode of naive concatenation is severe feature space entanglement and subsequent semantic erosion. Concatenating representations drawn from fundamentally different latent spaces forces the downstream distance metric to operate within a poorly conditioned, highly heterogeneous vector space. In this environment, features with naturally high magnitudes or high variances will disproportionately dominate the distance calculations (such as Euclidean or Cosine distance), entirely masking the complementary, discriminative signals of lower-magnitude features.[18] Furthermore, without explicitly learned cross-modal interactions, naive fusion invariably creates redundant, highly correlated subspaces. This redundancy wastes the representational capacity of the embedding vector and leads to suboptimal clustering dynamics during metric learning. A static concatenation or fixed scalar weighting cannot dynamically prioritize features based on the specific semantic context of the query image, meaning the system cannot adaptively rely on structural features when texture is ambiguous, or vice versa.[17]

Dataset Selection: Stress-Testing Generalization via Severe Domain Shift

Evaluating an image retrieval system exclusively on datasets that share similar underlying statistical distributions completely fails to assess the true generalization capacity, representation robustness, and metric space stability of the architecture. The provided assignment baseline mandates the utilization of CIFAR-10 and MNIST.[1] Both datasets, while foundational, present highly constrained and predictable challenges. MNIST consists of simple, stroke-based grayscale images with virtually no background noise, minimal intra-class variance, and effortlessly separable manifolds.[1] CIFAR-10 provides natural, object-centric color images with standard horizontal perspective projections, though constrained to an exceptionally low resolution of 32x32 pixels.[20]

To rigorously stress-test the proposed next-generation retrieval architecture, a third dataset must be selected that maximizes distribution diversity, introduces distinct and opposing inductive challenges, and simulates severe out-of-distribution domain shifts.

Selected Dataset: EuroSAT

The EuroSAT dataset is explicitly selected as the mandatory third evaluation benchmark for this research.[1] EuroSAT comprises 27,000 labeled and precisely geo-referenced satellite images captured by the Sentinel-2 satellite, categorized into 10 distinct land use and land cover classes.[22] While the dataset contains multispectral bands, the RGB subset provides a

sufficiently complex domain shift for standard visual architectures.

First-Principles Justification for EuroSAT

The selection of the EuroSAT dataset is rigorously justified through the analytical lens of data distribution variance, the required architectural inductive biases, and cross-domain generalization stress-testing:

1. Orthogonal Data Distribution and Perspective Invariance: Unlike CIFAR-10, where objects are systematically photographed with a horizontal perspective and standard canonical orientations (e.g., a car resting on the ground with the sky above), EuroSAT remote sensing images are captured from a strict top-down, nadir perspective.[24] Consequently, the geographic and structural objects within EuroSAT completely lack a canonical "up" or "down" orientation. This distribution fundamentally breaks the translation and rotation biases implicitly learned and relied upon by standard CNNs trained on natural images. The inclusion of EuroSAT forces the retrieval system to prove that it relies on absolute rotational invariance and robust topological feature extraction rather than memorized spatial orientations.[24]

2. Texture and High-Frequency Dominance over Bounded Shape: In natural image datasets like CIFAR-10, clear boundary delineations and convex shapes (e.g., the specific outline of a horse or an airplane) primarily dictate semantic meaning. Conversely, in the remote sensing domain of EuroSAT, classes such as "Annual Crop," "Forest," "Herbaceous Vegetation," or "Pasture" do not possess bounded, easily definable geometric shapes. Instead, their semantic identity is defined almost entirely by complex micro-textural patterns, color distributions, and high-frequency spatial repetitions extending to the borders of the image.[22] This provides an mathematically ideal testbed to validate whether the proposed novel feature representations can adequately capture frequency and structural texture when traditional bounded shape cues are entirely absent.

3. Severe Out-of-Distribution Domain Shift: Modern foundational vision models and standard CNN architectures are heavily biased toward the statistical properties of web-scraped natural images.[26] Applying a model optimized on the natural objects of CIFAR-10 directly to the satellite imagery of EuroSAT simulates a severe out-of-distribution domain shift.[16] A hybrid retrieval architecture that can maintain a high Mean Average Precision (mAP) across the stroke geometry of MNIST, the perspective objects of CIFAR-10, and the nadir textures of EuroSAT successfully demonstrates true domain-agnostic metric learning and representation robustness.[29]

Dataset Statistical Characteristics	**MNIST**	**CIFAR-10**	**EuroSAT**

Domain Classification	Handwritten Digits	Natural Objects	Remote Sensing / Satellite
Camera Perspective	2D Stroke Projection	Horizontal / Perspective	Top-Down (Nadir)
Primary Semantic Cue	Geometric Topology	Bounded Shape and Color	Texture and Spatial Frequency
Background Variance	None (Uniform black)	High (Natural unconstrained scenes)	High (Complex geographic terrain)
Resolution Constraints	28×28 (Grayscale)	32×32 (RGB)	64×64 (RGB / Multispectral)

Novel Feature Representation: Differentiable Spatial-Frequency Morphology (DSFM)

To directly address the diagnosed limitations of standard convolutional embeddings—specifically their inherent loss of high-frequency details, their rigid texture-shape entanglement, and their vulnerability to domain shifts—a completely novel feature representation is required.[1] The proposed representation, titled **Differentiable Spatial-Frequency Morphology (DSFM)**, mathematically unifies spectral frequency analysis, spatial texture extraction, and structural topology preservation into a single, end-to-end differentiable module.

The DSFM representation operates through two parallel, mathematically rigorous paradigms that subsequently interact: a Learnable Gabor Frequency manifold and a Probabilistic Morphological structure manifold.

Part 1: Learnable Gabor Frequency Extraction

Traditional spectral frequency transforms (such as the Discrete Wavelet Transform) and classical handcrafted Gabor filters require domain experts to manually define, tune, and hardcode filter scales, frequencies, and orientations prior to deployment.[31] DSFM resolves this rigidity by introducing a specialized convolutional layer where the spatial weights are strictly constrained to follow the mathematical formulation of a two-dimensional Gabor function, but the specific parameters governing the function are learned dynamically via backpropagation during the training phase.[33]

The 2D Gabor filter kernel $\Psi(x, y)$ is mathematically defined as a Gaussian envelope modulated by a complex sinusoidal plane wave:

$$\Psi(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(2\pi \frac{x'}{\lambda} + \psi\right)$$

Where the spatial coordinates are rotated by the orientation parameter θ:

$$x' = x \cos\theta + y \sin\theta$$

$$y' = -x \sin\theta + y \cos\theta$$

Instead of blindly learning an arbitrary dense $M \times M$ weight matrix with millions of unconstrained parameters like a standard CNN, the DSFM network learns only the compact parameter vector $\Omega = [\lambda, \theta, \psi, \sigma, \gamma]$ for a bank of K distinct filters.[33]

- λ **(Wavelength):** Dynamically captures the fundamental frequency of the target texture, adapting to varying spatial scales.
- θ **(Orientation):** Adapts to the statistically dominant angles of edges and textures in the specific dataset.
- σ **(Gaussian variance):** Explicitly controls the effective receptive field size of the filter without requiring deeper layers.
- γ **(Spatial aspect ratio):** Manages the ellipticity of the filter to capture elongated versus circular patterns.

Because the Gabor function is mathematically smooth and continuous with respect to all

variables in the parameter vector Ω, the gradients (e.g., $\frac{\partial \mathcal{L}}{\partial \lambda}$, $\frac{\partial \mathcal{L}}{\partial \theta}$) can be computed analytically and exactly via the chain rule during backpropagation.[35] This allows the frequency extractors to perfectly and optimally adapt to the specific texture distributions of diverse datasets like CIFAR-10 or EuroSAT without any human intervention.[36] The output of this stage is a highly parameterized, high-resolution frequency feature map denoted as F_{freq}.

Part 2: Differentiable Probabilistic Morphology (SoftMorph)

While the learnable Gabor filters excel at capturing fine-grained texture and spectral frequency, robust object retrieval also requires an explicit understanding of structural topology. Classical mathematical morphology (operations such as Erosion, Dilation, and Skeletonization) perfectly captures invariant shape topology. However, traditional morphology relies on discrete Boolean operations and non-differentiable min/max logic evaluated over a defined structuring element, which completely blocks gradient flow in deep neural networks.[37]

To resolve this, the DSFM module integrates a structural cue extractor utilizing the SoftMorph framework, which elegantly converts binary morphological operations into differentiable, multi-linear polynomials.[37]

Given a binary or thresholded input activation map Y where each pixel $Y_n \in $, a structuring element defines a precise neighborhood of k surrounding variables. A classical morphological **Erosion** (E) requires all k pixels within the defined neighborhood to be active:

$$E_{binary} = \bigwedge_{i=1}^{k} Y_i$$

Conversely, a classical **Dilation** (D) requires at least one pixel in the neighborhood to be active:

$$D_{binary} = \bigvee_{i=1}^{k} Y_i$$

To render these operations fully differentiable for representation learning, DSFM utilizes the quasi-probabilistic expectation of these operators by mapping Boolean expressions to

continuous polynomials using Fuzzy logic norms [37]:

$$E_{soft} = \prod_{i=1}^{k} Y_i$$

$$D_{soft} = 1 - \prod_{i=1}^{k} (1 - Y_i)$$

By sequentially cascading the E_{soft} and D_{soft} layers, the network formulates fully differentiable Opening (for noise removal) and Closing (for hole filling) operations.

Furthermore, DSFM implements a differentiable **Skeletonization (Thinning)** operator. Skeletonization preserves the topological centerline of an object regardless of its visual scale, texture, or geometric deformation.[37] Using 8-neighborhood sub-iterations mapped to soft polynomials, the network processes the image to yield a highly robust structural feature map F_{struct} that is fundamentally invariant to boundary noise and textural shifting.[37]

Feature Unification and the Novel Inductive Bias

The final DSFM feature representation is computed by integrating the outputs of the two paradigms through a depth-wise cross-attention mechanism that unifies the topological structural map with the spectral frequency map:

$$f_{DSFM} = \text{Attention}(Q = F_{struct}, K = F_{freq}, V = F_{freq})$$

By utilizing the structural skeleton map (F_{struct}) as the attention Query, and the Gabor frequency map (F_{freq}) as the Keys and Values, this mathematical formulation forces the network to sample high-frequency texture information *only* along the topologically significant structural skeletons of the image. This produces a unified feature vector f_{DSFM} that contains a fundamentally new and highly potent inductive bias: **topology-guided frequency extraction**.[1] This bias ensures that irrelevant background textures are suppressed, and only the textures belonging to the core structural geometry of the object are encoded for retrieval.

Hybrid Retrieval Architecture: Orthogonal Subspace

Adaptive Gating (OSAG)

Task 5 mandates the creation of a hybrid model designed to fuse standard CNN embeddings (f_{CNN}) with the proposed novel features (f_{DSFM}).[1] As diagnosed in the failure modes analysis, standard concatenation ($f_i =$) creates heavily entangled, heterogeneous feature spaces that suffer from catastrophic semantic erosion during distance calculations.[1]

To overcome this, the proposed architecture introduces **Orthogonal Subspace Adaptive Gating (OSAG)**. This advanced retrieval architecture explicitly models complementary feature subspaces using Gram-Schmidt orthogonalization and employs a sophisticated, retrieval-aware multi-objective loss function to guarantee manifold alignment and feature independence.[18]

1. Orthogonal Subspace Projection (OSP)

Both a deep ResNet-based CNN and the DSFM module will invariably capture a certain degree of redundant low-level information, such as basic edge gradients or primary color distributions. To maximize feature complementarity and ensure that the latent space is optimally utilized, the DSFM feature vector must be mathematically forced to represent *only* the specific structural and frequency information that is not already captured by the CNN semantics.[18]

Let $f_{CNN} \in \mathbb{R}^d$ be the invariant visual context generated by the deep convolutional network. Let $f_{DSFM} \in \mathbb{R}^d$ be the raw spatial-frequency feature extracted by the novel module. The OSAG architecture applies an Orthogonal Subspace Projection mechanism to rigorously decouple these features.[18] The network projects f_{DSFM} directly onto the orthogonal complement (the mathematical null space) of the semantic vector f_{CNN} :

$$f_\perp = f_{DSFM} - \mathrm{Proj}_{f_{CNN}}(f_{DSFM})$$

$$f_\perp = f_{DSFM} - \left(\frac{f_{DSFM}^T f_{CNN}}{\| f_{CNN} \|_2^2} \right) f_{CNN}$$

In this formulation, $f_\perp$ represents the purely orthogonal, decoupled modification increment.[18] Mathematically, the dot product $\langle f_{CNN}, f_\perp \rangle$ is strictly forced to 0. This geometric projection guarantees that the high-frequency and structural skeleton features provided by the DSFM

module are completely independent from the spatial and semantic features produced by the CNN. This eliminates representation redundancy, prevents high-magnitude CNN features from overshadowing nuanced textural cues, and entirely mitigates semantic erosion during metric comparison.[18]

2. Learned Adaptive Gating Fusion

Instead of relying on a fixed, hyperparameter-tuned scalar weight λ to balance the two features [1], the optimal reliance on frequency versus semantic features varies drastically on a per-image basis. When retrieving a specific car in CIFAR-10, the deep semantic shapes of f_{CNN} are highly vital. Conversely, when retrieving specific vegetation in EuroSAT, the texture and frequency features contained within $f_{\perp}$ are strictly paramount.[12]

To address this dynamic requirement, the OSAG architecture introduces a non-linear adaptive gating mechanism to fluidly and dynamically weight the orthogonal subspaces based on the exact contextual content of the input query image.[44] The network learns a dense projection matrix W_g that computes a context-aware attention weight vector $w \in^{d}$:

$$ w = \sigma \left(W_g [f_{CNN} |

| f_{\perp}] + b_g \right) $$

Where σ represents the sigmoid activation function. The final hybrid retrieval embedding f_{hybrid} is then calculated via the channel-wise Hadamard product ($\odot$):

$$f_{hybrid} = w \odot f_{CNN} + (1 - w) \odot f_{\perp}$$

This formulation guarantees a learned, highly adaptive fusion process where the neural network dynamically selects the optimal subspace vector on a per-channel, per-image basis, ensuring that the retrieval system can fluidly transition between semantic reliance and texture reliance depending on the specific domain of the query.[44]

3. Retrieval-Aware Optimization Pipeline

To effectively optimize this complex hybrid architecture specifically for the task of image retrieval, standard classification objectives such as Cross-Entropy loss are fundamentally insufficient, as they do not directly optimize the geometric distance metric between specific instances in the latent space.[47] Therefore, the OSAG architecture is trained end-to-end utilizing

a highly specialized joint objective function:

$$\mathcal{L}_{total} = \mathcal{L}_{triplet} + \alpha\mathcal{L}_{ortho} + \beta\mathcal{L}_{ctx}$$

1. Triplet Margin Loss ($\mathcal{L}_{triplet}$): This function directly optimizes the retrieval metric space by ensuring the Euclidean distance between an anchor image x_a and a positive image x_p (a relevant image) is strictly smaller than the distance between the anchor and a mined hard-negative image x_n by a defined minimum margin m.[47]

$$\mathcal{L}_{triplet} = \max(0, \|f_{hybrid}(x_a) - f_{hybrid}(x_p)\|_2^2 - \|f_{hybrid}(x_a) - f_{hybri}$$

2. Orthogonality Regularization Constraint ($\mathcal{L}_{ortho}$): To ensure that the mathematical independence of the feature subspaces is maintained continuously during the chaotic mini-batch gradient update process, an orthogonality penalty is applied. This enforces strict decorrelation between the CNN and DSFM feature spaces.[42]

\mathcal{L}_{ortho} = \mathbb{E} \left

3. Context Preservation Loss ($\mathcal{L}_{ctx}$): Inspired by techniques in compositional image retrieval, this geometric loss ensures that the final fused representation does not drift irreparably far from the primary, reliable semantic manifold established by the CNN baseline.[18] It strictly anchors the visual context.

$$\mathcal{L}_{ctx} = \left\| \frac{f_{hybrid}^T f_{CNN}}{\|f_{CNN}\|_2^2} f_{CNN} - f_{CNN} \right\|_2^2$$

Conceptual Architecture Diagram

The integration of these theoretical components can be visualized in the following conceptual data flow pipeline, illustrating the pathway from raw pixels to the final orthogonal hybrid embedding.

[Input Image Query: I]

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├──────────────────────────────────────────────────────┐
▼ ▼
(ResNet/ViT extracting (Extracting texture & topology)
spatial semantics) │
│ ├─> ──> F_freq
│ │
│ └─> ────> F_struct
│ │
│ ▼
│ [Cross-Attention Unification]
│ (Topology-guided frequency map)
▼ ▼
[f_CNN] ──────────────────────────┐
│ │ │
│ ▼ ▼
│
│ (Removes redundant semantics from DSFM)
│ │
│ ▼
│ [f_perp]
│ │
├──────────────────────────────────────┤
▼ ▼
──────┤
```

(Computes dynamic attention 'w') │

│ │

▼ ▼

[w ⊙ f_CNN] [(1 - w) ⊙ f_perp]

│ │

└──────────────────┬──────────────────┘

▼

[Hadamard Fusion]

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▼

[Final Fused Embedding: f_hybrid]

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Evaluation Strategy and Expected Behavior

A rigorous and mathematically exhaustive evaluation protocol is established to validate the theoretical claims of the DSFM representation and the OSAG hybrid architecture across the diverse domains of MNIST, CIFAR-10, and EuroSAT.[1]

Optimization Metrics for Retrieval Quality

The retrieval performance of the architecture will be exclusively quantified using standard, high-fidelity information retrieval metrics to provide a granular view of the embedding space geometry [1]:

- **Precision@K:** Evaluates exact local accuracy by computing the ratio of relevant images retrieved within the strictly top-K ranked results.
- **Recall@K:** Evaluates the capacity of the model to cluster the entire class manifold by computing the ratio of relevant images retrieved in the top-K against the absolute total number of relevant images in the database.
- **Mean Average Precision (mAP):** Serves as the primary optimization metric. It calculates the mean of the Average Precision scores across all queries, effectively integrating precision and recall into a single metric that reflects the overall ranking quality across the

entire recall spectrum.[1]

Expected Intra-Dataset Behavior

Based on the highly divergent statistical properties of the chosen datasets and the specific mathematical formulations of the respective models, the following comparative behavior is expected during intra-dataset retrieval analysis:

Evaluation Method	MNIST Expected mAP	CIFAR-10 Expected mAP	EuroSAT Expected mAP	Theoretical Robustness Remarks
LBP (Classical)	Moderate	Low	Moderate	Expected to fail entirely on the complex semantic shapes of CIFAR-10, but survive on EuroSAT's repetitive textures. Highly sensitive to background noise and varying resolutions.[1]
Shallow NN / MLP	High	Very Low	Very Low	Expected to memorize the simple 2D stroke grids of MNIST perfectly, but collapse completely on the perspective and texture shifts of

				CIFAR-10 and EuroSAT due to an absolute lack of spatial inductive bias.[8]
CNN Embeddings	High	High	Moderate	Expected to excel at identifying the distinct object semantics in CIFAR-10. However, struggles on EuroSAT because its inherent texture bias misaligns with multi-spectral, shape-absent OOD remote sensing features.[11]
Color Features	Low	Moderate	Low	Fails identically on MNIST (grayscale). Performs remarkably poorly on EuroSAT due to severe inter-class color overlap (e.g., all forests, pastures, and crops are variations of green, rendering

				color metrics useless).[1]
Proposed (DSFM)	Moderate	Moderate	**High**	The Learnable Gabor and SoftMorph modules will heavily dominate on EuroSAT's high-frequenc y, top-down textural structures. However, it lacks the deep, hierarchical semantic abstractions required to perfect CIFAR-10 retrieval independently.[33]
Hybrid (OSAG)	**Highest**	**Highest**	**Highest**	The adaptive gating network will dynamically select deep semantic CNN features when evaluating CIFAR-10 and seamlessly pivot to orthogonal DSFM frequency features for EuroSAT,

				allowing the model to dominate all single-paradigm baselines.[45]

(Note: Data distribution behaviors are synthesized from cross-domain generalization benchmarks and established representation learning literature.[8])

Cross-Dataset Zero-Shot Generalization Experiments

To truly and rigorously test representation invariance and evaluate the stress-testing capacity of the architecture, the models must be subjected to a Zero-Shot Cross-Domain evaluation protocol.[54]

1. **Experimental Protocol:** The baseline CNN and the OSAG hybrid model will be trained exclusively on the CIFAR-10 training dataset. Following training, all model weights will be rigidly frozen. The models will then be forced to extract embeddings for the EuroSAT dataset to execute queries and database ranking.
2. **Theoretical Hypothesis:** Standard CNN architectures will experience a catastrophic mAP collapse (e.g., dropping from an 85% mAP on CIFAR-10 to below 30% on EuroSAT) because the models have severely overfitted to horizontal perspective shapes and canonical orientations.[53] The OSAG architecture, however, is hypothesized to maintain robust, generalized performance. The adaptive gating module will mathematically detect the out-of-distribution semantic collapse of the f_{CNN} branch representations. Consequently, the network will automatically route higher activation weights to the $f_{\perp}$ (DSFM) branch, relying almost entirely on the domain-agnostic Gabor frequencies and invariant morphological skeletons to successfully execute the retrieval task without prior training on satellite imagery.[46]

Ablation and Failure Case Analysis

A comprehensive ablation study is required to isolate and empirically prove the contribution of each distinct branch in the architecture. Disabling the Orthogonal Subspace Projection (thereby reverting the system to naive vector concatenation) will definitively demonstrate the negative mathematical impact of semantic erosion, evidenced by a significant drop in mAP whenever the representations of f_{CNN} and f_{DSFM} conflict in magnitude.[18] Disabling the Gabor frequency branch will cause severe accuracy degradation exclusively on EuroSAT, while disabling the SoftMorph branch will cause the model to fail on the structural topologies of MNIST, proving that the combined features are strictly complementary.[37]

Theoretical Failure Modes of the Hybrid OSAG System: Despite its robust mathematical foundations, the OSAG architecture remains theoretically vulnerable in highly specific, adversarial edge cases. For instance, if an image dataset features fine-grained classes where the core semantic identity is *entirely* and intrinsically correlated with a specific texture (e.g., a dataset specifically designed to separate "woven wooden baskets" from "knitted wool baskets"), the Gram-Schmidt orthogonalization process may act too aggressively. In projecting the DSFM vector onto the null space of the CNN, the projection might aggressively strip away the texture variance from the CNN feature space, pushing that vital discriminative information entirely into the independent DSFM space.[18] If the adaptive gate fails to perfectly balance this severe split due to a poorly initialized W_g matrix, the rigid orthogonality constraint ($\mathcal{L}_{ortho}$) could inadvertently suppress highly discriminative but correlated features, leading to a slight reduction in Precision at exactly K=1.[42] Furthermore, extreme environmental lighting conditions that cast deep shadows could obscure the mathematical structuring elements required by the SoftMorph skeletonization process, temporarily blinding the topological extraction branch and forcing the network to rely solely on the fallback CNN semantics. Addressing these specific edge cases provides the immediate trajectory for future research and optimization within the OSAG paradigm.

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